

unexpected. So you can see here that the Black Swan is a sucker's problem. In other words, it occurs relative to your expectation. You realize that you can eliminate a Black Swan by science (if you're able), or by keeping an open mind. Of course, like the LTCM people, you can create Black Swans with science, by giving people confidence that the Black Swan cannot happen—this is when science turns normal citizens into suckers.

Note that these events do not have to be *instantaneous* surprises. Some of the historical fractures I mention in [Chapter 1](#) have lasted a few decades, like, say, the computer that brought consequential effects on society without its invasion of our lives being noticeable from day to day. Some Black Swans can come from the slow building up of incremental changes in the same direction, as with books that sell large amounts over years, never showing up on the bestseller lists, or from technologies that creep up on us slowly, but surely. Likewise, the growth of Nasdaq stocks in the late 1990s took a few years—but the growth would seem sharper if you were to plot it on a long historical line. Matters should be seen on some relative, not absolute, timescale: earthquakes last minutes, 9/11 lasted hours, but historical changes and technological implementations are Black Swans that can take decades. In general, positive Black Swans take time to show their effect while negative ones happen very quickly—it is much easier and much faster to destroy than to build. (During the Lebanese war, my parents' house in Amioun and my grandfather's house in a nearby village were destroyed in just a few hours, dynamited by my grandfather's enemies who controlled the area. It took seven thousand times longer—two years—to rebuild them. This asymmetry in timescales explains the difficulty in reversing time.)

A BRIEF HISTORY OF THE BLACK SWAN PROBLEM

This turkey problem (a.k.a. the problem of induction) is a very old one, but for some reason it is likely to be called “Hume's problem” by your local philosophy professor.

People imagine us skeptics and empiricists to be morose, paranoid, and tortured in our private lives, which may be the exact opposite of what history (and my private experience) reports. Like many of the skeptics I hang around with, Hume was jovial and a bon vivant, eager for literary fame, salon company, and pleasant conversation. His life was not devoid of anecdotes. He once fell into a swamp near the house he was building in Edinburgh. Owing to his reputation among the locals as an atheist, a woman refused to pull him out of it until he recited the Lord's Prayer and the Belief, which, being practical-minded, he did. But not before he argued with her about whether Christians were obligated to help their enemies. Hume looked